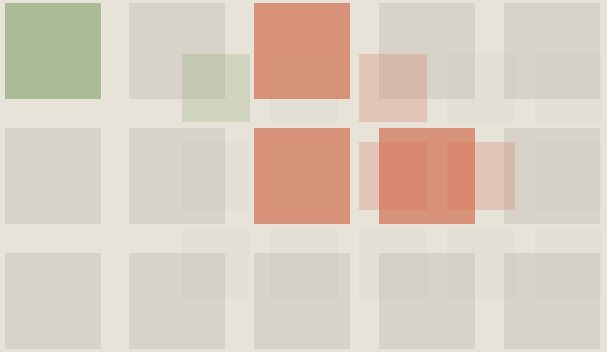
Anomaly detection for data quality, measured inside the ETL pipeline

Living progress deck. Rule-based, classical and first deep detectors run through one pipeline and are compared per anomaly type, on a synthetic generator and on 42 real tables with injected anomalies, and on 100 labelled problems of the pool.



Author	Theo Bigand · updated 2026-10-21
Repository	github.com/theobigand/etl-anomaly-benchmark
Code	1,150 tests · 100% line and branch coverage · 97.75% mutants killed
Datasets	pool of 152 open sets · Credit Card, Cardio, Water Quality · synthetic, 6 anomaly types

02 THE QUESTION

Does machine learning in an ETL pipeline beat plain rules, and for which anomalies?

The industry default is rule-based: auditable, cheap, and blind to what the author did not anticipate. Recent work adds ML detectors inside the flow, but the evidence is thin, one algorithm on one dataset. This thesis builds the missing head-to-head benchmark and breaks the result down by anomaly type.

What generic anomaly benchmarks leave out

EVALUATED IN A VACUUM

Algorithms scored outside any pipeline, no integration cost measured.

NO DECLARATIVE BASELINE

The rules that dominate production are never in the comparison.

ONE AGGREGATE SCORE PER DATASET

Hiding which family is good at which kind of defect.

A SINGLE FLAG THRESHOLD

Reported as one operating point, not the precision-recall trade-off.

03 THE ORIGINAL CONTRIBUTION

A controlled dataset that labels six anomaly types separately

12,360 records generated with Faker and NumPy, six types injected at 3% each, every row tagged with its category. Recall **and** precision are reported per type. The magnitudes are one **severity** multiplier, so the choice is swept rather than fixed.

6

anomaly types, labelled and measured separately

MISSING VALUE	one or two feature cells set to null
EXACT DUPLICATE	a clean row copied, same business attributes
UNIVARIATE OUTLIER	one numeric field pushed far out of range
MULTIVARIATE OUTLIER	every field in range, the combination not (an 18-year-old with a top-percentile balance)
TEMPORAL DRIFT	recent-signup rows whose income and balance distribution has shifted up
INVALID FORMAT	malformed e-mail, out-of-domain country code, impossible age

Because the category of every injected row is known, the benchmark reports a recall **and** a precision per type: the part that generic anomaly benchmarks, which publish one aggregate score, do not provide.

04 THE SYSTEM

One interface, one measured path

The demo and every benchmark call the same `detect()` stage, so latency is the real in-pipeline fit + predict + score time. A continuous `score()` on top of the boolean `predict()` makes ROC-AUC and PR-AUC computable; `detect(fit_df=...)` fits on the train slice and scores the test set.

extract → transform → **detect** → load

```python
# every detector, one contract
class Detector(ABC):
    name: str
    def fit(self, df) -> Self: ...
    def predict(self, df) -> Series:   # bool per row
        ...
    def score(self, df) -> Series:     # graded score
        return self.predict(df).astype(float)

# classical ML: adapt PyOD, do not reimplement
class PyODDetector(Detector):
    def fit(self, df):
        x = impute(df[self.numeric_cols])
        self.model = self.estimator.fit(scale(x))
        return self

# registry: one line, picked up everywhere
"isolation_forest": lambda: IsolationForestDetector(),
```

05 TWO FAMILIES NOW, A THIRD PLANNED

Seven detectors, one contract

Classical ML detectors are wrapped from PyOD, not reimplemented. The flag threshold is read from the train-set prevalence. The time-aware `drift` detector is opt-in and runs on the synthetic dataset only. Deep learning (Autoencoder, LSTM-Autoencoder) is M2.

Detector	Family	Principle	Main limit
Z-score	statistical rule	flag `|z| > 3` on any column	mean and std inflated by outliers
IQR	statistical rule	Tukey fences, `k = 1.5`	any-column rule fires often in high dim
Great Expectations	declarative rule	not-null + range at 0.1 / 99.9 pct	only checks what the suite declares
Isolation Forest	classical ML	isolation depth under random splits	axis-aligned splits, weak on nulls
Local Outlier Factor	classical ML	local density vs 20 neighbours	scoring cost grows with the scored set
One-Class SVM	classical ML	RBF boundary around the bulk	quadratic training
drift	time-aware	recent-window mean shift, per-row lean	per-row, so diluted drift is invisible

06 FOUR DATASETS, ONE PROTOCOL

Three real settings bracket the method

Stratified 70 / 15 / 15 split, fit on train, every number reported on the held-out test set. Five seeds, mean ± std. The ML contamination is read from the train-set prevalence, not fixed at 10%.

0.01 → 0.20

Isolation Forest fraud precision, from the contamination change alone

Dataset	Rows	Feat.	Anom.	Role
Credit Card Fraud	283,726	30	0.17%	rare, but a mixed-defect label
Cardio	1,831	21	9.6%	rare and outlying, the fitting regime
Water Quality	3,276	9	61%	target class common, wrong regime
Synthetic per-type	12,360	9	17.5%	controlled, 6 labelled types

LOF and One-Class SVM run everywhere except Credit Card Fraud, where neither scales to 284k rows in the pipeline.

07 HOW EACH TYPE IS INJECTED

Magnitudes are a swept parameter, not a fixed choice

severity = 1.0 is the reference dataset. Missing values, duplicates and invalid formats have no magnitude to scale. The severity sweep (slide 15) turns "we picked the magnitudes" into a curve and shows the per-type ranking is stable. Five types replace clean rows; duplicates are appended.

MISSING VALUE	1–2 of {income, balance, avg amount, tenure, n txn, email, country} → null
DUPLICATE	exact copy of a sampled clean row, id included
UNIVARIATE	one numeric column × U(15, 45) × severity
MULTIVARIATE	marginals at the (1 − 0.03 / severity) percentile of the clean data
TEMPORAL DRIFT	signup in the last 80 days; income × (1 + 0.55 · severity), similar for balance and amount
INVALID FORMAT	email → malformed; age → impossible; country → out of domain

At `severity = 1` these are the reference magnitudes. Sweeping the multiplier (slide 15) shows every distributional detector's recall on univariate, multivariate and drift rises smoothly, and the ranking does not change.

Aggregate metrics say the detector families are interchangeable. Per anomaly type, they are not.

F1 spans 0.31 to 0.51 on the synthetic set · the per-type competences barely overlap

The operating point decides the outcome

The statistical rules recover most frauds at precision below 0.04. Isolation Forest changes once its contamination matches the prevalence: it flags 0.17% of rows and reaches the first usable precision, at recall 0.21 and with real seed variance (F1 = 0.20 ± 0.07). Great Expectations leads PR-AUC.

0.20

Isolation Forest precision (was 0.01 at contamination 0.10)

0.182

Great Expectations PR-AUC, the best on this dataset

Detector	Prec	Rec	F1	ROC	PR	Flag%	Time
Z-score	0.011	0.870	0.021	0.926	0.078	13.4	0.03s
IQR	0.003	**0.966**	0.007	0.911	0.018	48.6	0.12s
Great Expectations	0.037	0.676	0.070	0.830	**0.182**	3.1	1.37s
Isolation Forest	**0.202**	0.206	**0.203**	**0.934**	0.120	0.2	1.78s

Mean over 5 seeds, 15% held-out test split. LOF and One-Class SVM do not scale to 284k rows here.

When anomalies are rare and outlying, the learned detectors work

Isolation Forest reaches F1 0.62 and PR-AUC 0.70. Z-score also does well: the 21 features are standardized and anomalies stand out on several at once. This is the counterpoint to a 0.17% mixed label and a 61% majority.

0.62
Isolation Forest F1 on Cardio

0.70
Isolation Forest PR-AUC on Cardio

Detector	Prec	Rec	F1	ROC	PR	Flag%
Z-score	0.49	**0.82**	0.61	0.93	0.61	16.6
IQR	0.21	0.90	0.34	0.78	0.30	43.4
Great Expectations	0.51	0.21	0.29	0.59	0.19	4.6
Isolation Forest	**0.64**	0.62	**0.62**	**0.95**	**0.70**	9.9
LOF	0.25	0.27	0.26	0.62	0.24	10.7
One-Class SVM	0.56	0.61	0.58	0.94	0.63	10.7

Cardiotocography, 1,831 rows, 9.6% anomalies. Mean over 5 seeds, test split. Wide seed spread on this small set (ML F1 std up to 0.17).

Aggregate metrics make the families look interchangeable

Isolation Forest has the best F1; no family is clearly ahead. Z-score keeps a perfect precision by firing only on the most extreme rows. One-Class SVM matches Isolation Forest at about 16× the latency.

0.31–0.51

F1 range across all six detectors

0.33–0.56

PR-AUC range: no clear winner

Detector	Prec	Rec	F1	ROC	PR	Time
RULE-BASED						
Z-score	**1.000**	0.182	0.308	**0.748**	**0.560**	0.002s
IQR	0.603	0.399	0.481	0.678	0.460	0.004s
Great Expectations	0.920	0.205	0.335	0.601	0.332	0.29s
CLASSICAL ML						
Isolation Forest	0.515	**0.503**	**0.509**	0.745	0.501	0.16s
Local Outlier Factor	0.380	0.390	0.385	0.646	0.437	0.14s
One-Class SVM	0.488	0.478	0.483	0.729	0.531	2.61s

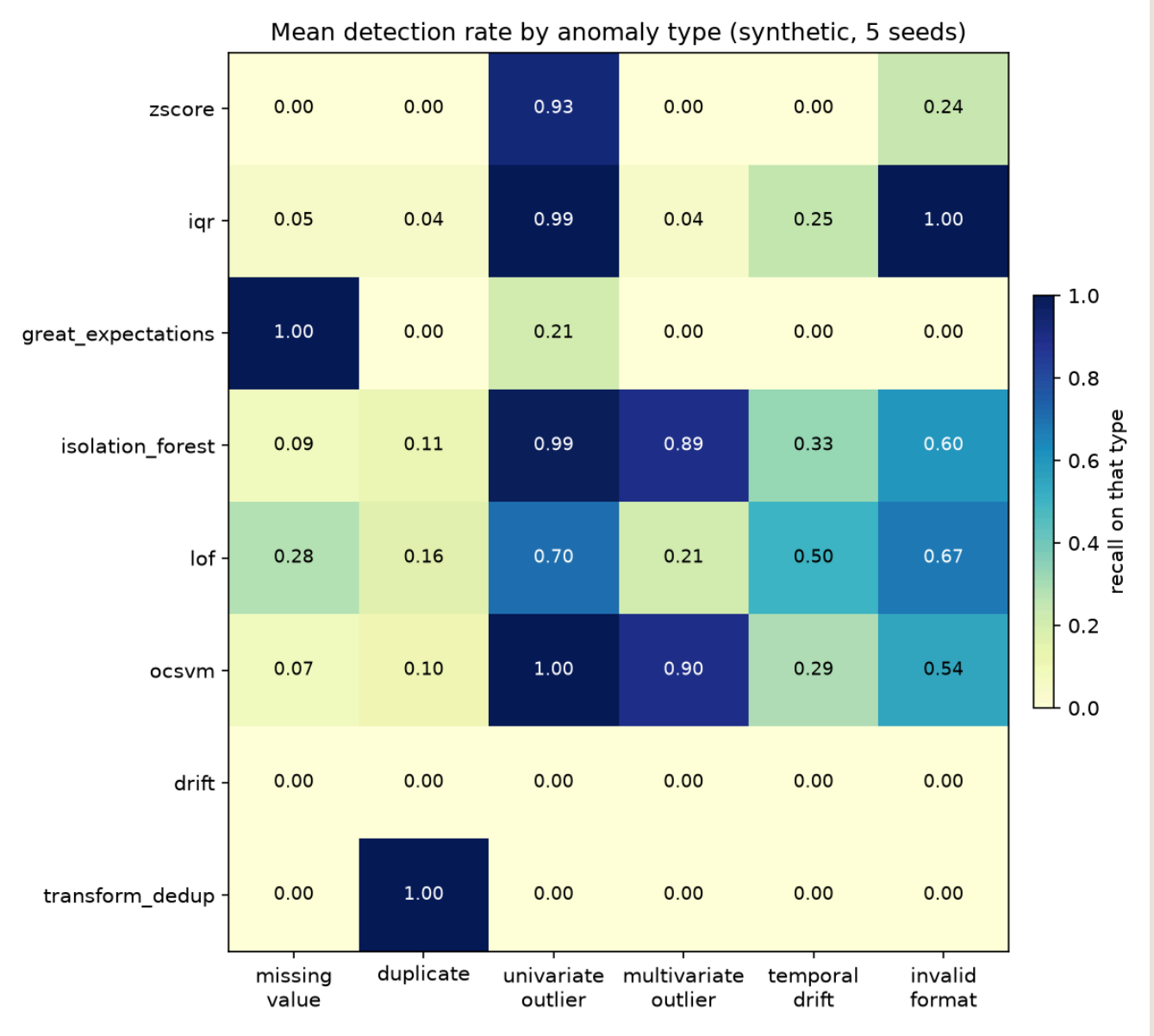
Mean over 5 seeds, test split. 12,360 rows, 17.5% anomalous.

Each method has a narrow competence

Z-score owns univariate outliers. Great Expectations owns missing values (recall 1.00, precision 0.90). Deduplication owns exact duplicates. Isolation Forest and One-Class SVM are the only methods that catch multivariate outliers. Broad coverage needs several families, not the single best one.

0.91 vs ≤0.02

Isolation Forest recall on multivariate outliers, against every rule



paper/figures/heatmap_synthetic.png

transform_dedup and drift shown for reference

13 RECALL AND PRECISION, PER TYPE

Detection rate alone can be gamed

Isolation Forest catches 91% of multivariate outliers at only 0.24 precision: a wide net that also pulls in clean rows. Z-score fires only when it is certain, so its precision is 1.00 on the two columns it covers.

0.91 / 0.24

recall / precision on multivariate outliers, Isolation Forest

Method	missing		univ. out.		multiv. out.		drift	
	rec	prec	rec	prec	rec	prec	rec	prec
Z-score	0.00	–	**0.92**	**1.00**	0.00	–	0.00	–
IQR	0.05	0.03	**0.99**	0.39	0.02	0.01	0.27	0.15
Great Expectations	**1.00**	**0.90**	0.23	0.70	0.00	–	0.00	–
Isolation Forest	0.09	0.03	**0.99**	0.26	**0.91**	0.24	0.33	0.10
LOF	0.26	0.06	0.65	0.15	0.16	0.04	**0.49**	0.11
One-Class SVM	0.08	0.03	**0.99**	0.25	0.84	0.22	0.34	0.10
transform dedup	0.00	–	0.00	–	0.00	–	0.00	–

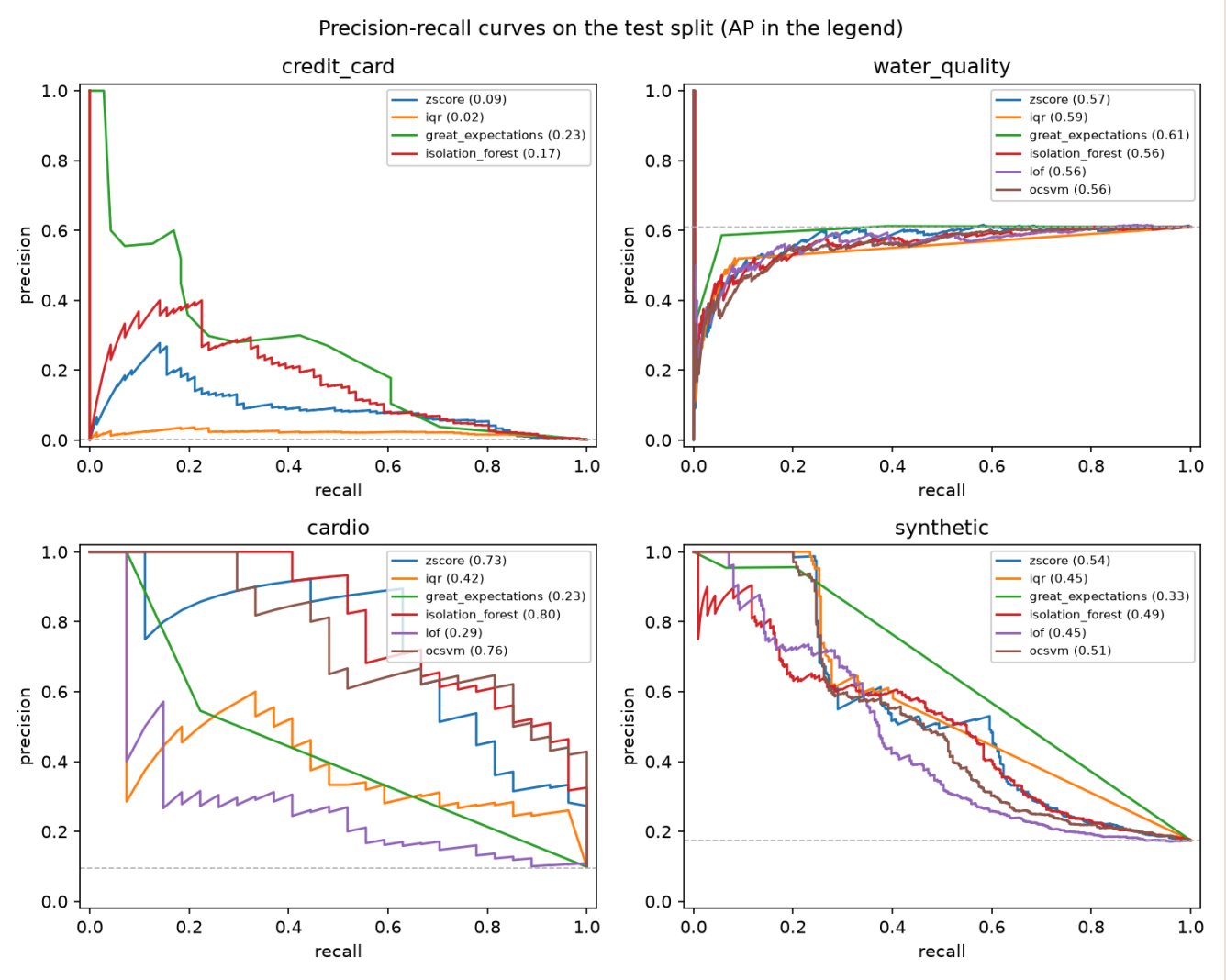
Recall and precision per anomaly type, clean rows as the only negatives. 5 seeds, mean. Duplicate and invalid-format columns omitted for space.

One threshold hides the whole trade-off

On Cardio the Isolation Forest and One-Class SVM curves stay high (AP 0.80, 0.76). On Water Quality every curve hugs the 0.61 base-rate line. On Credit Card Fraud two methods dominate but neither clears precision 0.4 except at negligible recall.

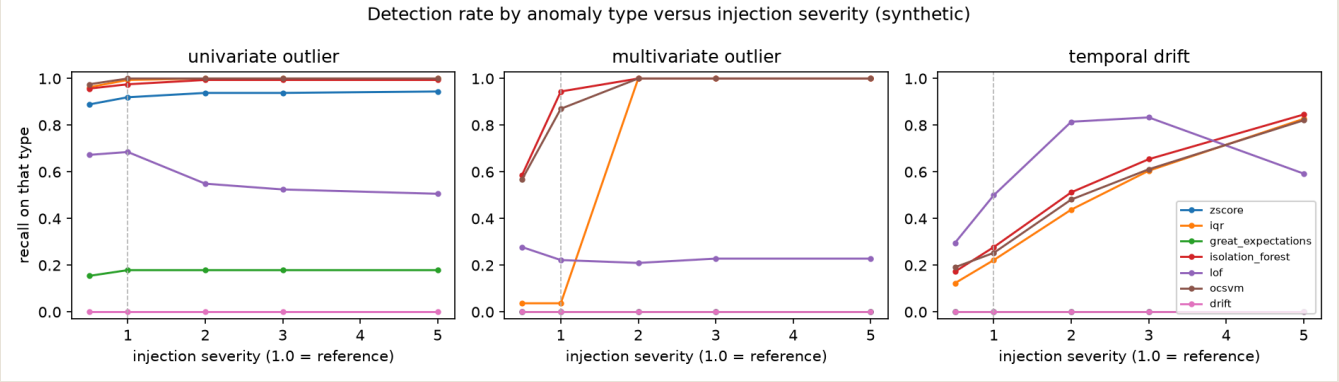
0.37 → 0.008

Isolation Forest fraud precision as recall goes 0.23 → 0.92



The per-type ranking is stable as the magnitudes scale

Univariate outliers are saturated already at severity 0.5. Multivariate outliers show a threshold. Temporal drift climbs roughly linearly for the distributional detectors: the reference drift is simply below every method's floor, not a property of the methods. The same families lead the same columns at every severity.



≤0.3 → 0.85

temporal-drift recall from severity 1 to 3

A pool of 152 open datasets, pinned and checked

Issue #66. The three datasets of M1 bracket the method but are too few to say anything general. The pool follows ADBench in scale and adds a licence rule: a dataset enters only when its terms are read and open. Raw data stays out of the repository; checksums, licences and reports are archived.

152
pinned datasets

116
clean carriers for injection

100
labelled anomaly problems

ADBench tabular sets kept (35 CC BY 4.0, 1 ODbL)	36 OF 47
UCI sets passing metadata and licence filters	133
UCI sets kept after curation, reason recorded	116
anomaly label rule fixed before any detector	RAREST CLASS
one normalisation policy for all sets	SINGLE
download check of every set, manual and monthly CI	PINNED
excluded for unclear terms (celeba, mammography, vowels…)	11

On 100 labelled problems, no detector is clearly ahead

Six detectors, default settings, five seeds, on the 36 ADBench and 64 UCI problems of the pool. Protocol written before the run. Learning helps a little: Isolation Forest beats the Z-score on 67 problems, by 0.01 to 0.04 of PR-AUC. Three of five hypotheses hold; the two that fail are reported as they are.

100
problems run, none failed

30
problems where no detector reaches ROC-AUC 0.6

Detector	Problems	PR-AUC median [95%]	ROC-AUC	Rank
One-Class SVM	84	**0.240 [0.177, 0.307]**	0.670	3
Isolation Forest	100	0.215 [0.158, 0.293]	**0.677**	**2**
Z-score	100	0.203 [0.164, 0.279]	0.626	3
LOF	92	0.165 [0.153, 0.237]	0.588	3
IQR	100	0.144 [0.117, 0.191]	0.556	4
Great Expectations	100	0.115 [0.105, 0.123]	0.508	5

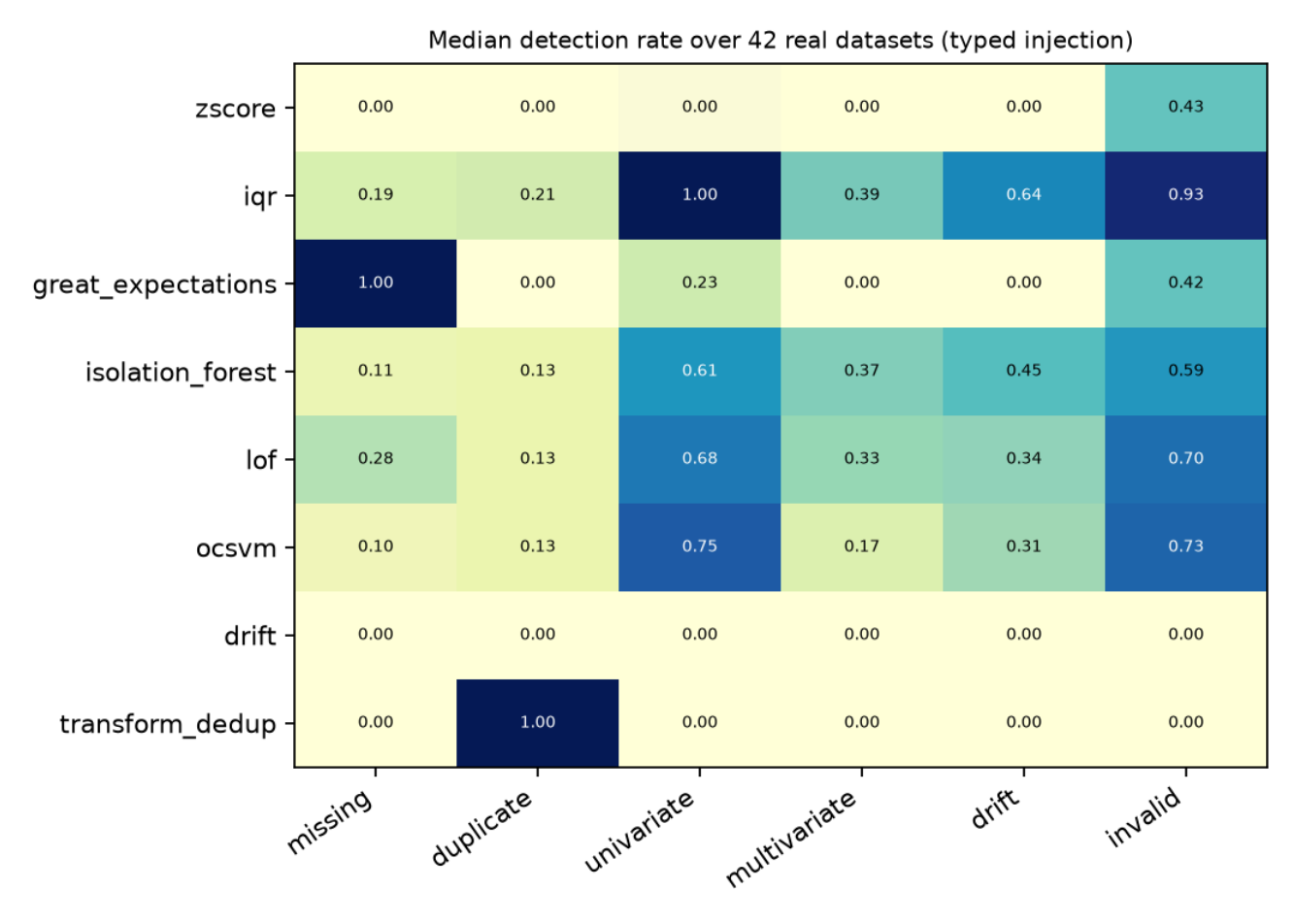
Median over problems. LOF and One-Class SVM are skipped on large sets, the skips are archived. The order of methods is the same for ADBench and UCI (rank correlation 0.94), the level is lower on UCI.

The synthetic pattern does not simply carry over

Issue #65. The six anomaly types are injected into 42 real UCI tables (3% of rows per type, same split and settings as the synthetic run). Protocol and five hypotheses were written before any run. Two rows hold everywhere: Great Expectations finds all missing values, deduplication all duplicates. Three hypotheses are not supported.

0.37
Isolation Forest, multivariate outliers (0.89 synthetic)

22%
median clean rows flagged by IQR



median over 42 tables, injected not natural

Two failures that are explained, not hidden

Z-score detects almost nothing on univariate outliers (median 0.00 against 0.93 synthetic). A check made after seeing it, so exploratory, shows why: invalid rows in the training split inflate the standard deviation. Without them the Z-score finds every univariate outlier on all 42 tables. The IQR fences reach the best detection rate on several types, but only by flagging a median 22% of clean rows.

0.83
mean rank correlation between two tables

-0.96
IF detection vs number of columns

MASKING

Non-robust statistics on a contaminated training set, not a weakness of the Z-score as such.

DETECTION RATE CAN BE GAMED

The false positive rate on clean rows has to be read next to it. It reaches 99% on some wide tables.

STABLE RANKING

Rank correlation of the method-by-type matrices is 0.83, lowest for drift (0.70). Partly mechanical: many cells sit near 0 or 1.

DIMENSION HURTS

Isolation Forest on univariate outliers falls as the number of continuous columns grows.

Time changes the answer: a threshold from training does not travel

Credit Card is the only dataset with a clock. Detectors are fitted on the first 70% of time and scored on the last 15% (52 frauds). Average precision drops for all four detectors, but the interval excludes the stratified mean only for Great Expectations and Isolation Forest. Prevalence also changes between partitions, so the pilot does not separate it from the effect of time.

Detector	PR strat.	PR chrono	ROC chrono	Flag%
Z-score	0.078	0.048	0.935	12.8
IQR	0.018	0.013	0.917	47.1
Great Expectations	0.183	0.045	0.758	3.5
Isolation Forest	0.110	0.035	0.936	0.08

Average precision on the test split, 95% block-bootstrap interval for the chronological arm in the paper. The window monitor detects mean shifts of 1 and 2 standard deviations, not 0.5.

0.936
Isolation Forest ROC-AUC, chronological

0.08%
future rows it flags: none of the 52 frauds

An LSTM autoencoder is comparable, not clearly better

Issue #52. The window is the ten previous rows of the global stream, since Credit Card has no cardholder. The protocol and a CPU budget of 20 epochs were fixed beforehand. On a synthetic stream with broken continuity the LSTM wins by a wide margin. On the real frauds it only matches Isolation Forest and Great Expectations, and none of five trainings converged within the budget.

0.875
LSTM AP on the synthetic stream (row AE 0.006)

0.037
LSTM AP on Credit Card (row AE 0.024)

paired bootstrap, main seed	+0.021 [+0.005, +0.044]
other four seeds	INTERVAL INCLUDES 0
versus Isolation Forest (0.035), GX (0.045)	COMPARABLE
convergence within 20 epochs	NOT REACHED
recall of frauds in the later period	≤ 0.02
deep models in the per-type matrix	#64 DONE

The profile is stable, two cells are not

Issue #67. Transactions and sensor readings get the same six injected types and the same protocol as the customer table. Great Expectations and the deduplication rule carry over exactly, and the rank correlation between schemas is 0.93. The multivariate column and drift depend on how the columns are scaled and correlated. Two of four hypotheses hold.

0.93
mean rank correlation between schemas

0.96
best drift recall on sensors (0.38 on transactions)

Multivariate outliers	customers	transactions	sensors
Isolation Forest	0.89	0.49	0.32
One-Class SVM	0.90	0.18	0.33
IQR	0.04	0.57	0.02
Z-score	0.00	0.00	0.00

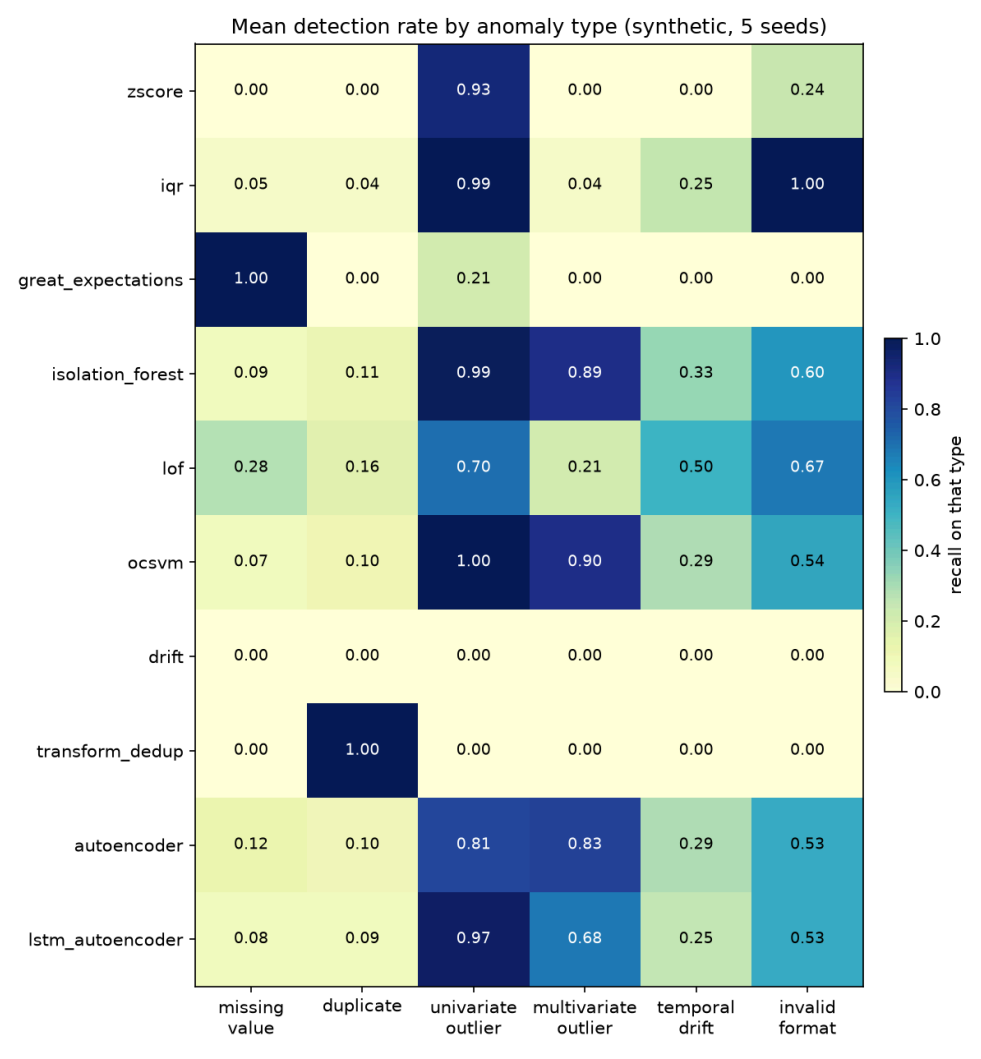
Mean recall over five seeds. Same generator logic for all three schemas: the agreement shows the profile does not depend on one table, not that it holds on real data.

A sequence model does not fix temporal drift

Issue #64. An autoencoder and an LSTM autoencoder, both trained to convergence (60 to 178 epochs, never the cap), added to the synthetic matrix on the same splits. Three of four hypotheses hold. The one that fails is the one I expected to matter: the LSTM catches 25% of drifted rows, LOF 50%.

0.25
LSTM recall on temporal drift (LOF 0.50)

0.83
autoencoder recall on multivariate outliers



Detection latency is not pipeline latency

Issue #53. Synthetic matrices from 1,000 to 300,000 rows, a fresh process per measurement so that peak memory belongs to that run, 300 s budget. On the full Credit Card file the Z-score pipeline takes 4.8 s of which detection is 3%, while Isolation Forest takes 8.5 s with detection at 43 to 45%.

133 s

LOF at 300,000 rows (exponent 1.51)

>300 s

One-Class SVM at 100,000 rows

Detector	10k rows	100k rows	300k rows	Memory MB
Z-score	0.002 s	0.013 s	0.039 s	70
IQR	0.004 s	0.029 s	0.086 s	116
Isolation Forest	0.16 s	1.09 s	3.2 s	140
Autoencoder (MLP)	1.5 s	8.4 s	12.7 s	115
LSTM autoencoder	1.1 s	5.1 s	7.3 s	499
LOF	0.73 s	18.0 s	133 s	405
One-Class SVM	5.6 s	>300 s	not run	788

Median detection-stage time, 8 features. LSTM trained for at most 5 epochs on 50,000 rows: a budget cap.

25 THE PRACTICAL OUTPUT

Anomaly type to method

Two of six types are not a per-row detector problem at all: duplicates belong to the transform stage, drift to a windowed monitor. That itself is a finding a single aggregate ranking would never surface.

Anomaly type	Use	Recall	Why
missing value	Great Expectations	1.00	not-null expectation, exact
exact duplicate	transform dedup stage	1.00	row equality, not a detector's job
univariate outlier	any statistical rule	0.92–0.99	single-column band is enough
multivariate outlier	Isolation Forest	0.91	only the joint models see it (precision 0.24)
temporal drift	windowed monitor	≤0.49	sub-threshold per row; needs a stage
invalid format	IQR, partially	1.00 / none	numeric part caught, string format missed

Unsupervised detection works in one of three regimes

The question is less "rules or ML" than "does this dataset have rare, outlying anomalies at all". Latency and scaling then constrain which detector can sit in a real flow: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

Fraud

too extreme

0.17% positives, and the label mixes every defect. No method reaches a usable precision at reasonable recall.

Cardio

the fit

9.6%, rare and distinct. Isolation Forest F1 0.62, PR-AUC 0.70. The regime the detectors are built for.

Water

wrong regime

61% majority target. Every ROC-AUC 0.45 to 0.51: the detectors are at chance.

The synthetic dataset sits in the middle at 17.5% and is where the per-type structure is measured. Pipeline cost narrows the shortlist further: Great Expectations is the slowest linear method, and the One-Class SVM and LOF do not survive the full fraud dataset.

Findings from the build

Each of these became a paragraph in the paper. The recurring lesson: the setup around a detector, the operating point, the regime, the pipeline cost, matters as much as the algorithm.

THE OPERATING POINT IS A LEVER

Contamination = prevalence instead of a fixed 10% moves Isolation Forest fraud precision from 0.01 to 0.20 and multivariate recall from 0.41 to 0.91.

ONE REGIME OUT OF THREE

Cardio: F1 0.62. Fraud (mixed label): nothing usable. Water Quality (61% majority): at chance.

GX 1.X IS A FULL API REWRITE

Per-row failures need the COMPLETE result format plus an explicit index column. Figured out by probing the installed library.

ONE-CLASS SVM DOES NOT SCALE

Quadratic training, and LOF scoring is superlinear. On the full fraud dataset only rules and Isolation Forest run.

ML IS WEAK ON MISSING VALUES

Imputing removes the signal a not-null rule catches for free: GX recall 1.00 against 0.08 to 0.26 for the learned models.

DRIFT IS NOT A PER-ROW PROBLEM

Once diluted by clean recent rows the window mean has moved by about a tenth of a standard deviation.

What the repository holds today

The verification layer was built for the claim that the numbers are right: a test suite, a mutation campaign, deliberate defects and a release script that returns certified, incomplete or blocked. The last local run is not independently attested, and the hosted run certified an earlier revision.

1,150
tests, plus 3 on full datasets

97.75%
mutants killed (4,700 of 4,808)

Check	Result	Scope
line coverage	1,628 / 1,628	pipeline modules and experiment scripts
branch coverage	336 / 336	same scope
mutation, computing code	95.76%	split, metrics, generator, detectors
mutation, input / output	98.87%	readers, writers, scripts
fault injection	23 / 23	four went through every earlier test
real-data check	pass	three CSV files, manuscript tables match

Survivors (108) count as failures and were classified by reading the changes, not by proof.

Experiments run, write-up and deep rows left

Five of the seven M2 plan items are not closed. The target is an IMRaD paper; the demo paper was dropped. Defense July 2027.

held-out split, 5 seeds, contamination, CI	DONE
chronological pilot, LSTM pilot, cost study	DONE
dataset pool of 152 sets (#66)	DONE
typed injection on 42 real tables (#65)	DONE
benchmark on the 100 labelled pool problems	DONE
deep rows in the per-type matrix (#64)	DONE
multi-schema Faker generator (#67)	DONE
related work, verified bibliography (#68)	NEXT
IMRaD restructuring, condensed verification (#69)	M2
decision-level study under temporal shift (#63)	M2

Rules and machine learning are complementary by anomaly type.

The per-type heatmap is what makes that choice actionable. Across three real datasets, unsupervised detection pays off only on Cardio, where the anomalies are genuinely rare outliers. On 100 labelled problems of the pool no detector is clearly ahead. On 42 real tables with injected types, the synthetic pattern holds only in part, which is why the real-data matrix is the differentiator. The pipeline, the 152-dataset pool, the protocols and the verification layer are in place. The IMRaD write-up and the remaining studies are what remains.